ST5202 — Applied Regression Analysis
Lecture Notes 7. Model Diagnostics

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Outline

Motivation and strategy

Systematic diagnostics

Local diagnostics

Example: surgical unit data

Summary

Why do we need model diagnostics?

- A fitted regression model is a set of assumptions imposed on the data.
- For a given data set, these assumptions are not necessarily satisfied exactly.
- Model diagnostics is used to detect important discrepancies between the fitted model and the data.
- The goal is not to prove that the model is true, but to assess whether it is adequate for inference.

Two kinds of diagnostic questions

A fitted model may show two kinds of discrepancy.

- Systematic discrepancy: the overall model structure is inadequate.
- Local discrepancy: a small number of observations behave unusually and may distort the fit.

In this lecture, we study these two aspects separately.

Linear model assumptions

Recall the linear model:

$$y_i = X_i\boldsymbol{\beta} + \epsilon_i, \quad \epsilon_i \overset{i.i.d.}{\sim} N(0, \sigma^2), 1 \leq i \leq n.$$

Assumptions:

- Linear relationship $X_i\boldsymbol{\beta}$
- Constant variance $\mathrm{Var}(\epsilon_i)$ for all samples $1 \leq i \leq n$
- Independent noise across samples
- Noise follow normal distribution
- All relevant predictors are included in the model

One or multiple of them may be violated

Residuals as the basic object for systematic diagnostics

- The fitted value of the ith observation is

$$\hat{y}_i = \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}}, \qquad i = 1, \ldots, n.$$

- The residual is

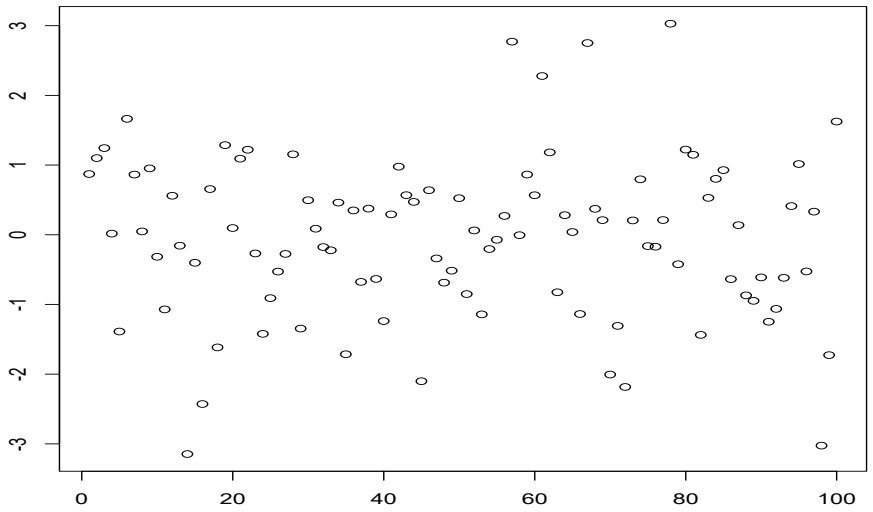
$$e_i = y_i - \hat{y}_i, \qquad i = 1, \ldots, n.$$

- Residuals represent the part of the response not explained by the fitted model.
- For systematic diagnostics, we mainly inspect the overall pattern of residuals.

Null pattern of residual plots

If the fitted model is broadly adequate, the residuals should show no systematic structure.

- centered around zero,
- roughly evenly scattered,
- no visible trend,
- no obvious change in spread.



Checking the mean structure

We check whether the linear relationship $X_i\beta$ adequately describes

$$E(Y \mid X).$$

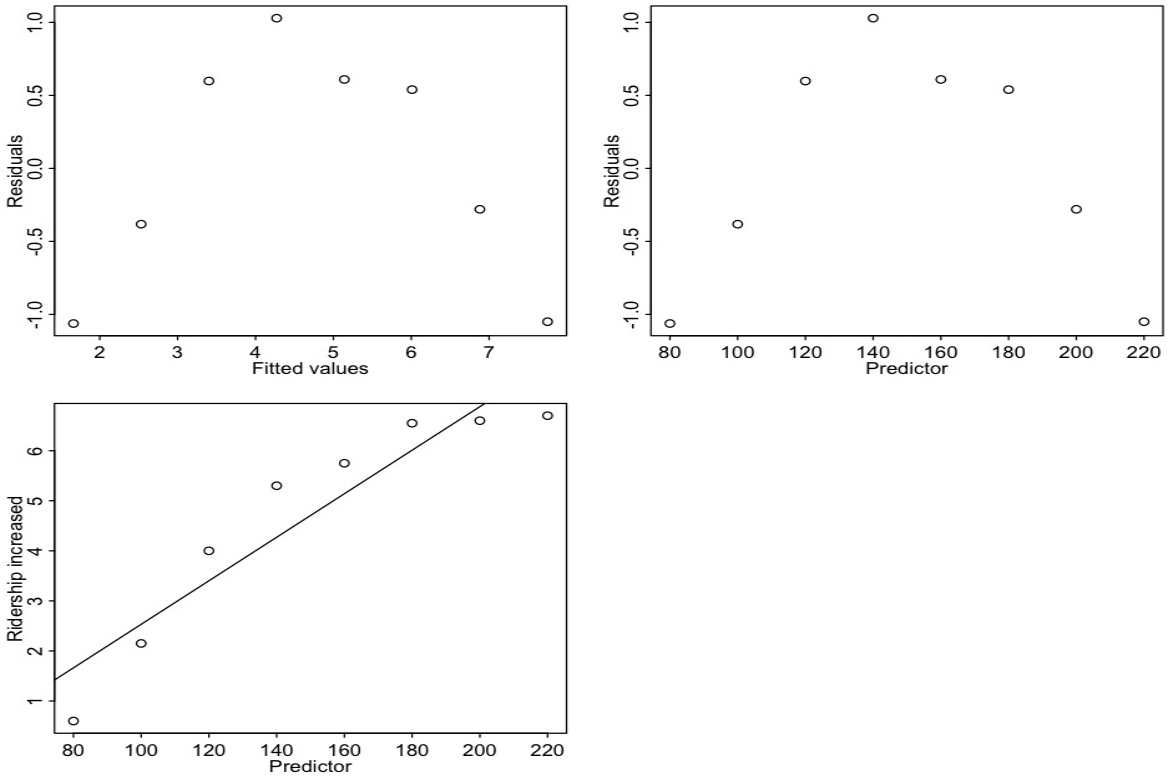
Useful plots include

- residuals against fitted values – global structure
- residuals against predictors (one plot each predictor) – unexplained part
- response against predictors (one plot each predictor) – marginal pattern

A curved trend or any systematic departure from zero suggests that the mean structure may be misspecified.

Example: misspecified mean structure

If a clear trend remains in the residual plot, the regression function may be misspecified.



Checking constant variance

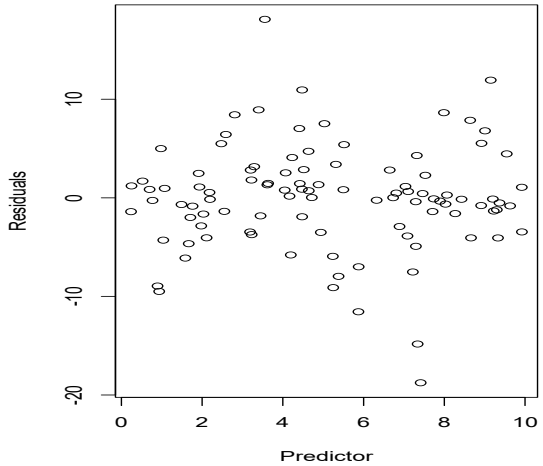
We check whether the error variance is approximately constant.

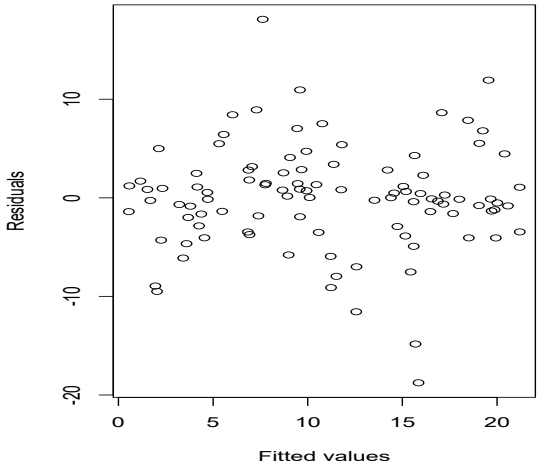
- Residual plots against fitted values and predictor variables are useful.
- If the variance is not constant, the vertical spread of residuals will change along the horizontal axis.
- Typical patterns include funnel shapes and systematic widening or narrowing of the spread.

Non-constant variance mainly affects the reliability of standard errors and inference.

Example: non-constant variance

A changing spread of residuals indicates possible heteroscedasticity.





Checking independence

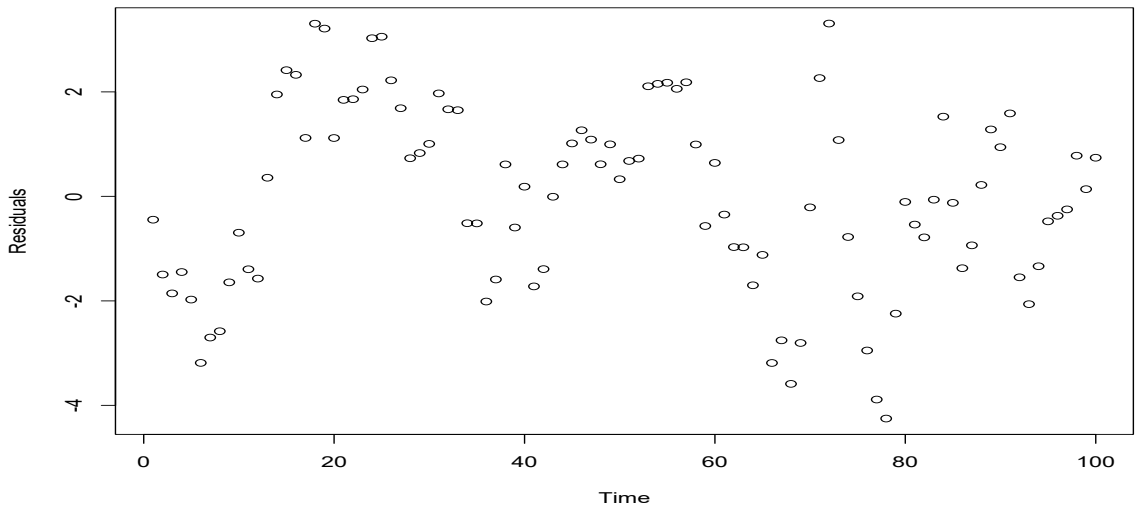
Independence is checked by plotting residuals against the order in which the observations are obtained, such as

- time,
- space,
- sampling order.

Runs, cycles, clustering, or visible dependence among adjacent residuals suggest lack of independence.

Example: non-independence

The following residual plot shows a visible dependence pattern over adjacent observations.



Studentized residuals

We introduce the studentized residuals to check normality

- Consider the linear model $Y = X\beta + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I_n)$. Let
$$H = X(X^\top X)^{-1}X^\top$$
be the hat matrix, and let h_{ii} be its ith diagonal element, i.e.
$$h_{ii} = \boldsymbol{x}_i^\top (X^\top X)^{-1}\boldsymbol{x}_i,$$
which is called the leverage value of the ith observation.
- For the residual $e_i = y_i - \hat{y}_i$, the variances differ as
$$\mathrm{Var}(e_i) = \sigma^2(1 - h_{ii}),$$
- To standardize the residuals, we define the studentized residual
$$r_i' = \frac{e_i}{\hat{\sigma}\sqrt{1 - h_{ii}}}, \qquad i = 1, \ldots, n.$$

Check normality by studentized residuals

The normality can be checked by the plots of studentized residuals.

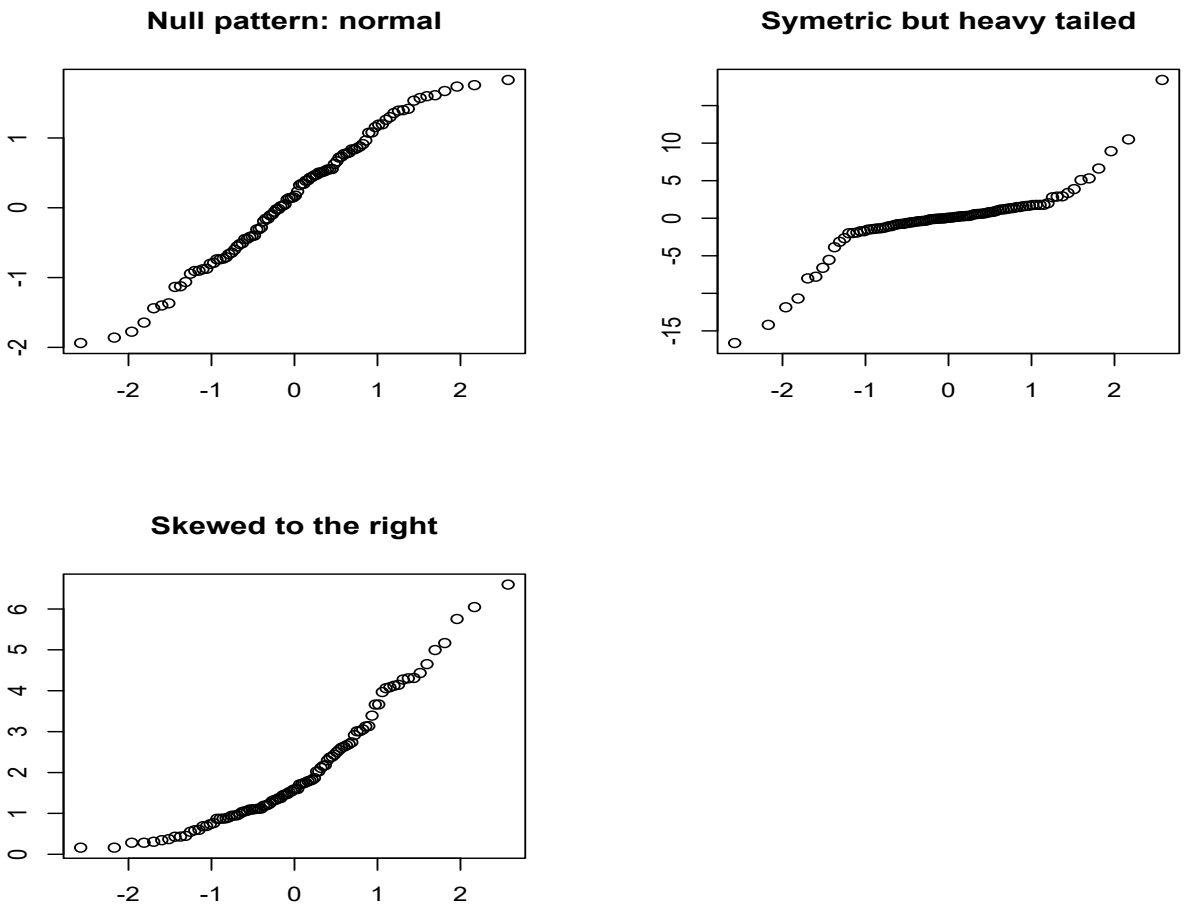
Common graphical tools include:

- histogram of the studentized residuals;
- boxplot of the studentized residuals;
- normal probability plot or Q-Q plot of the studentized residuals.

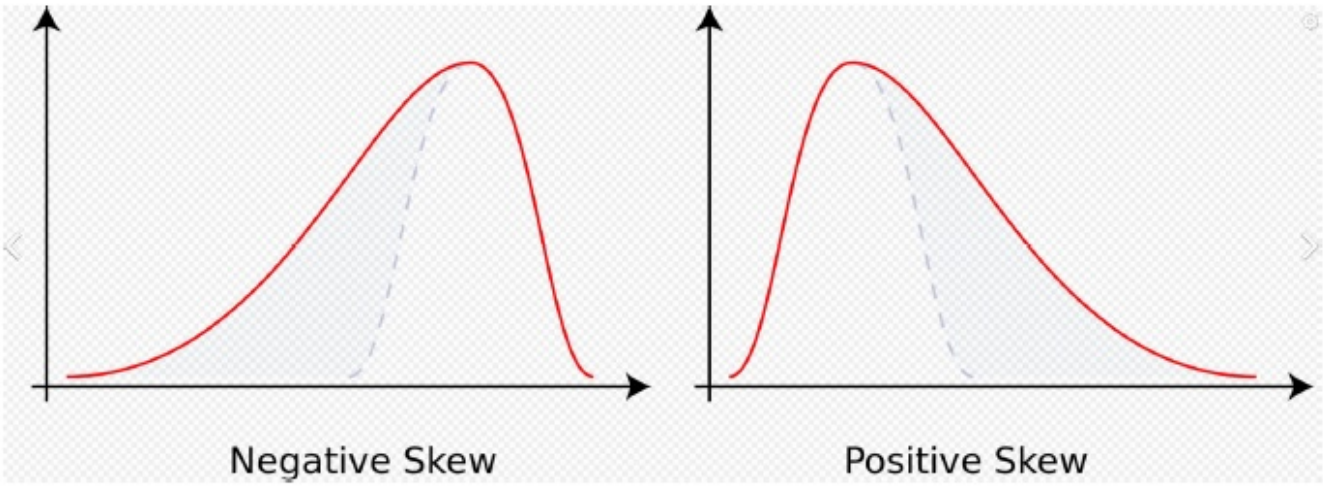
If the normality assumption is reasonable, then the points in the Q-Q plot should lie approximately on a straight line.

Systematic departures from the straight line may indicate non-normality, such as skewness or heavy tails.

Example: Q-Q plot of residuals



Skewness



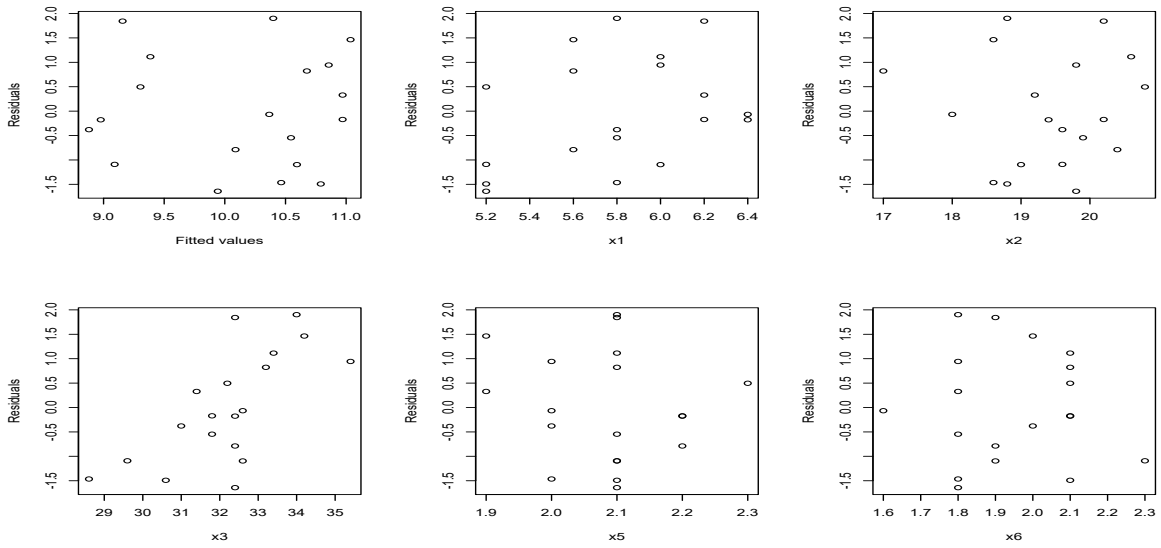
A distribution is right-skewed (or positively skewed) if its right tail is longer than its left tail.

A distribution is left-skewed (or negatively skewed) if its left tail is longer than its right tail.

Residual plots can reveal omitted predictors

Residual plots against predictors not included in the model may reveal omitted predictors.

- In the following example, the model is fitted only to X_4, and the residual plot suggests that X_3 is a missing predictor.



Summary of systematic diagnostics

Residual plots are used to assess whether the fitted model shows systematic inadequacy in

- the mean structure (linear assumption),
- the variance structure,
- independence,
- normality (using studentized residual).

If no clear systematic pattern is present, the model passes these checks reasonably well.

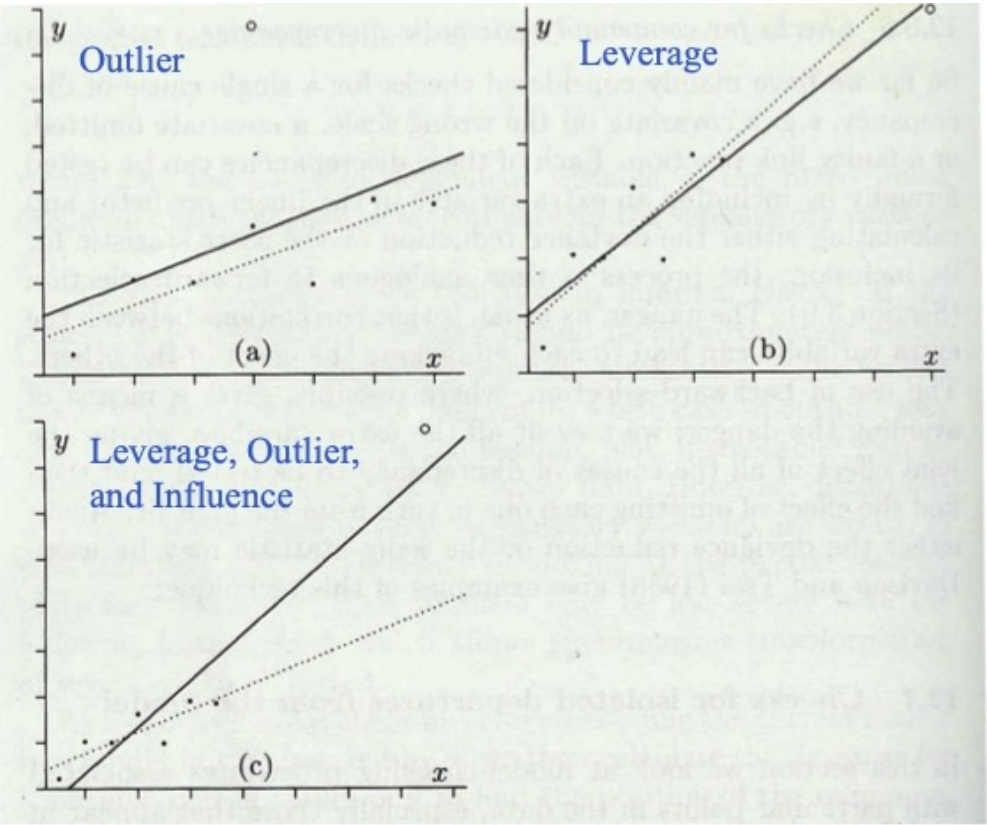
Unusual observations: three different notions

When individual observations are examined, three notions should be distinguished.

- *Leverage point*: unusual in the predictor space.
- *Outlier*: unusual in the response direction.
- *Influential point*: has a substantial effect on the fitted model.

These notions are related, but not equivalent.

Outlier, leverage, and influence together



Leverage and hat values

- Recall that we have the hat matrix $H = X(X^\top X)^{-1}X^\top$ and the hat value $h_{ii} = \boldsymbol{x}_i^\top (X^\top X)^{-1}\boldsymbol{x}_i$ as the i-th diagonal of H.
- A large h_{ii} indicates that observation i is far from the bulk of the data in the predictor space.
- Since $\sum_{i=1}^n h_{ii} = \mathrm{tr}(H) = p$, a common empirical rule is to flag observations with

$$\text{flag:} \quad h_{ii} > \frac{2p}{n}.$$

Outlier and residuals

To detect outliers, we use standardized residual-based quantities.

- Studentized residuals

$$r_i' = \frac{e_i}{\hat{\sigma}\sqrt{1-h_{ii}}}.$$

- Studentized deleted residuals

$$r_i^* = \frac{y_i - \hat{y}_{i(i)}}{\hat{\sigma}_{(i)}\sqrt{1-h_{ii}}} = \frac{y_i - \hat{y}_i}{\hat{\sigma}_{(i)}\sqrt{1-h_{ii}}}.$$

  where $\hat{y}_{i(i)}$ and $\hat{\sigma}_i$ correspond to the predicted value at X_i and estimated sd when the ith observation is deleted.

- A large $|r_i^*|$ suggests that observation i is unusual in the response direction.

Influence and Cook's distance

Influence concerns how much the fitted model changes when an observation is removed.

- Cook's distance

$$d_i = (\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}})^\top (X^\top X)(\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}})/(p\hat{\sigma}^2),$$

  where $\hat{\boldsymbol{\beta}}_{(i)}$ is the estimate of $\boldsymbol{\beta}$ when the ith observation is deleted.
- A large d_i indicates that deleting observation i changes the fitted model substantially.
- Such observations deserve closer examination.

Screening unusual observations in practice

Useful quantities for screening unusual observations include

- hat values h_{ii},
- studentized deleted residuals r_i^*,
- Cook's distances d_i.

A practical strategy is to inspect plots or ordered values of these measures, and then flag unusually large observations for closer examination.

R functions for local diagnostic measures

For a fitted linear regression object `fit`, the commonly used diagnostic quantities can be obtained by

```
fitted.values(fit)        # fitted values
residuals(fit)            # ordinary residuals
hatvalues(fit)            # leverage values
rstandard(fit)            # studentized residuals
rstudent(fit)             # studentized deleted residuals
cooks.distance(fit)       # Cook's distances
```

These quantities are useful for screening unusual observations.

Assessing candidate outliers

A candidate outlier should be investigated rather than automatically removed.

- To assess observation i, introduce the dummy variable

$$u = \begin{cases} 1, & \text{for the } i\text{th unit,} \\ 0, & \text{otherwise.} \end{cases}$$

- The significance of the coefficient of u indicates that the corresponding observation is inconsistent with the fitted model.
- If several observations are assessed simultaneously, multiplicity should be taken into account, for example by Bonferroni adjustment.

Surgical unit data

A hospital surgical unit was interested in predicting survival in patients undergoing a particular liver operation. A random sample of 54 patients was available for analysis.
For each patient, the following predictors were recorded.

X_1: blood clotting score,

X_2: prognostic index,

X_3: enzyme function test score,

X_4: liver function test score.

The response variable is log survival time.

The data set is provided in the file *surgical.txt* on Canvas.

R code for the surgical unit model

The model can be fitted by

```
fitSurg <- lm(log.survival ~ clotting + prognostic + enzyme
+liver, data = surgical.dat)
summary(fitSurg)
```

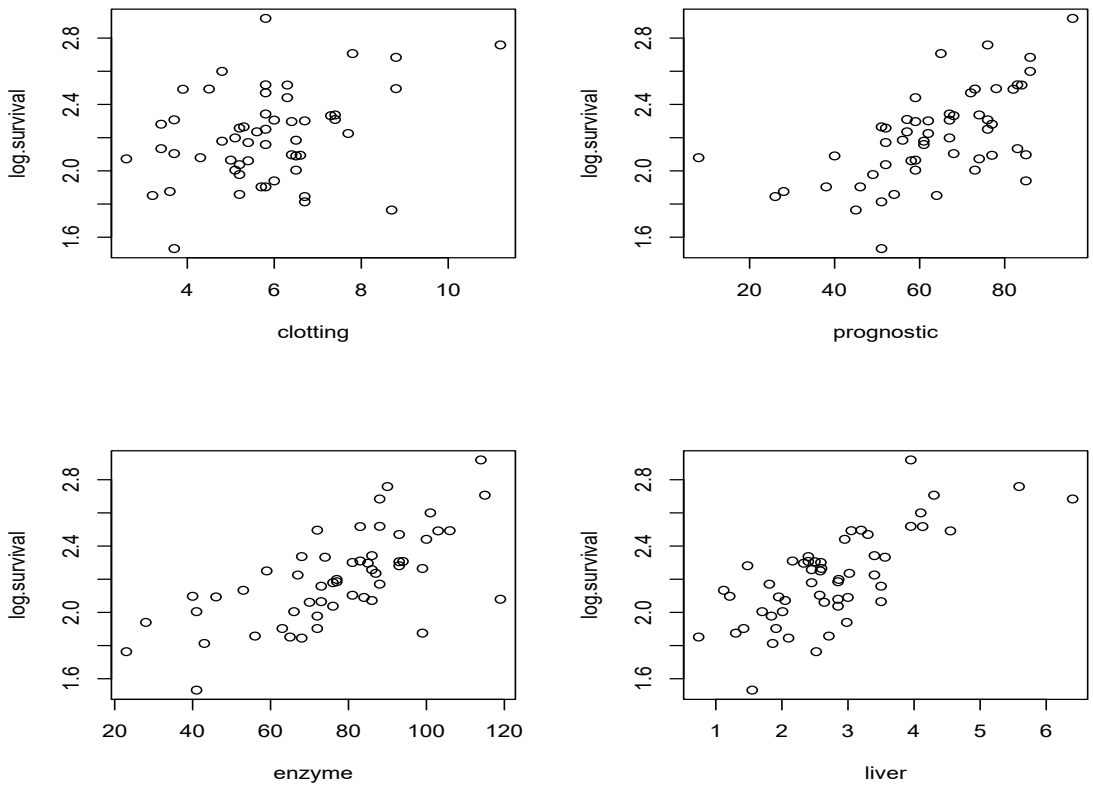
The basic diagnostic quantities are then obtained by

```
lev  <- hatvalues(fitSurg)
delr <- rstudent(fitSurg)
cook <- cooks.distance(fitSurg)
```

Analysis strategy for the surgical unit data

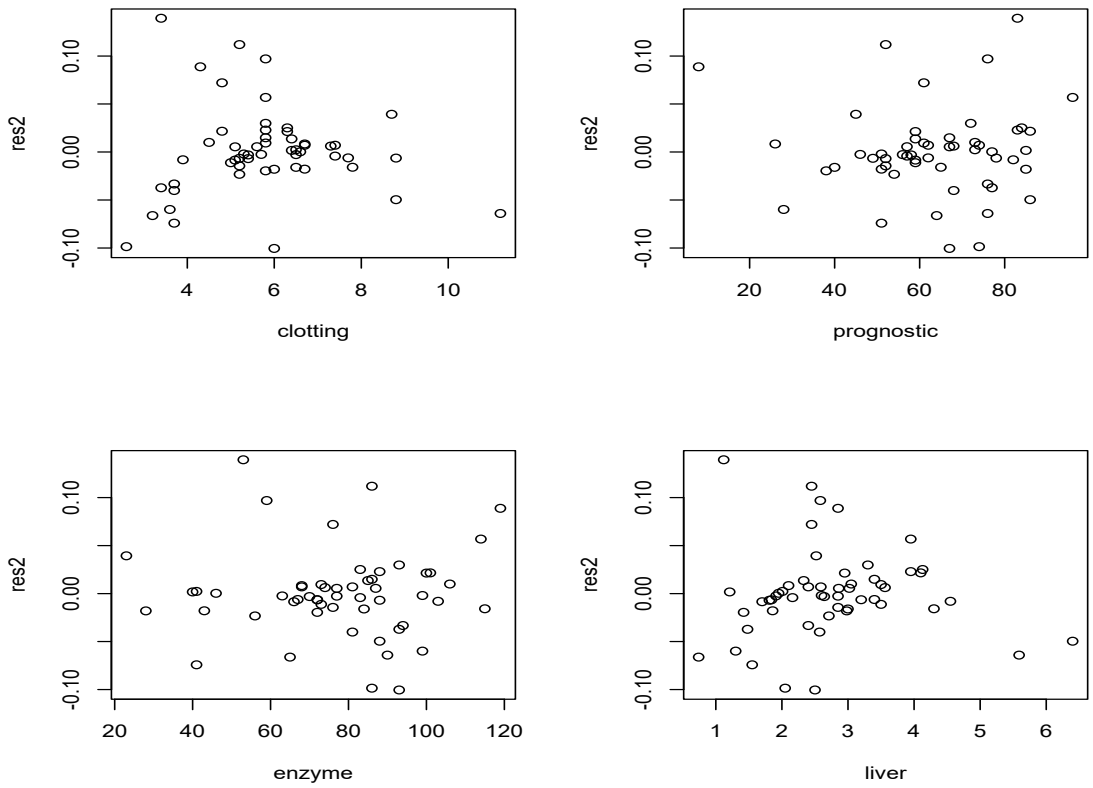
- Scatter plots of the response against the predictors suggest approximately linear relations.
- The full model is selected by the forward procedure.
- We use both systematic diagnostics and local diagnostics to assess the adequacy of the fitted model.

Checking the linearity: plots of $\log(Y)$ vs. predictors



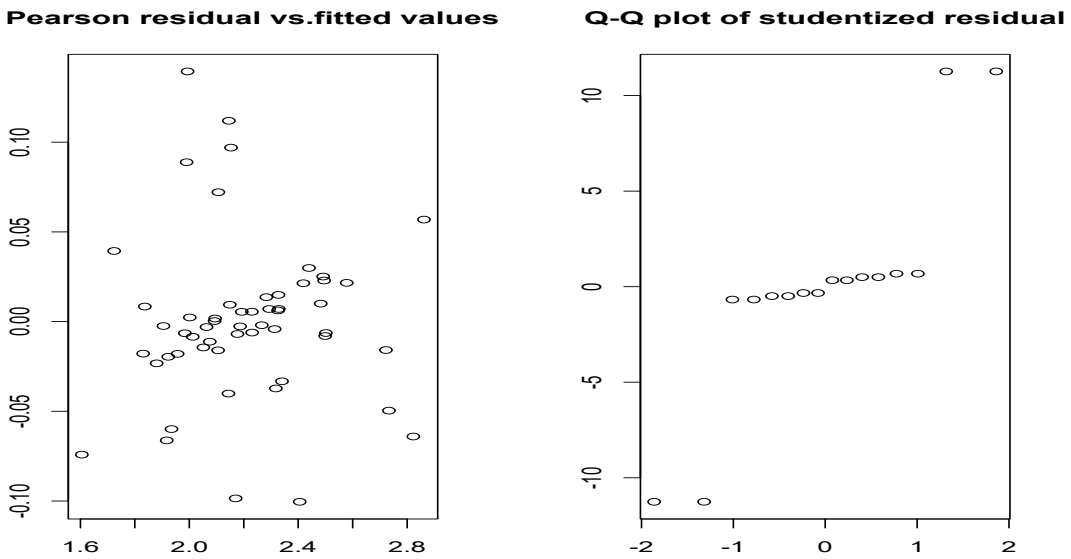
The plots suggest approximately linear relations between the response and the selected predictors.

Checking linearity: residual plots against predictors



The residual plots do not show obvious trends, indicating no major discrepancy in the regression function.

Checking variance and normality



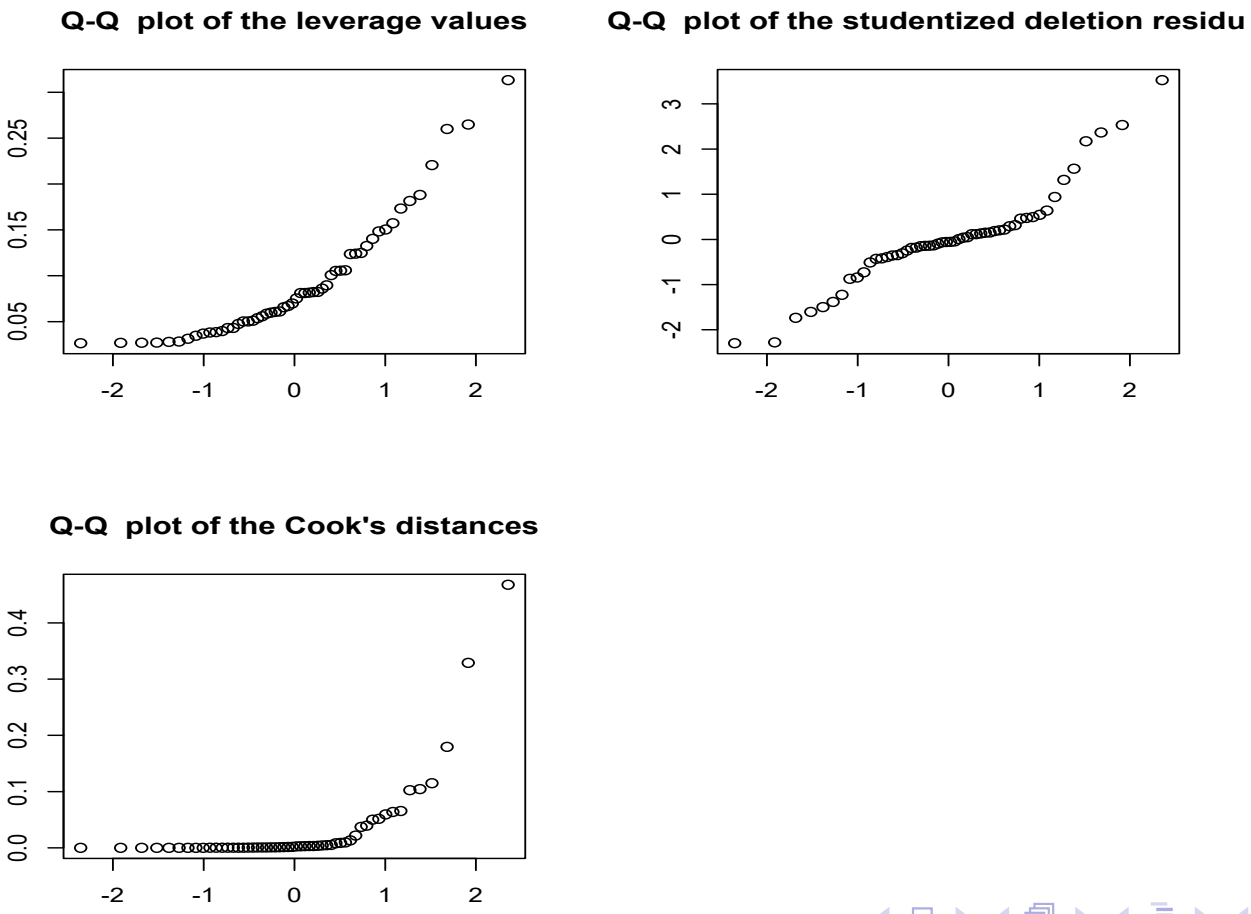
(1) There is no strong evidence of non-constant variance.
(2) The Q-Q plot suggests possible heavy tails, which may indicate non-normality or unusual observations.

Screening unusual observations

For local diagnostics, we inspect leverage, studentized deleted residuals, and Cook's distance.

- observations with large h_{ii} may be high-leverage points;
- observations with large $|r_i^*|$ may be outliers in the response direction;
- observations with large d_i may be influential.

Q-Q plots of local diagnostic measures



Screening unusual observations in R

High leverage points can be screened by

```
p <- length(coef(fitSurg))
which(lev > 2*p/nrow(SurgUnit))
```

Observations with the largest deleted studentized residuals can be found by

```
sort(abs(delr), decreasing = TRUE)[1:4]
```

Observations with the largest Cook's distances can be found by

```
sort(cook, decreasing = TRUE)[1:4]
```

Candidate unusual observations

- Observations with $h_{ii} > 2p/n$:

```
    13     17     18     28     32     38     42
0.1572 0.1880 0.1505 0.2649 0.2206 0.3132 0.1483
    43     50     52
0.2599 0.1816 0.1733
```

- Observations with the four largest absolute studentized deleted residuals:

```
       37         9        27        38
-2.296534 -2.279002  2.531081  2.368453
```

- Observations with the four largest Cook's distances:

```
        37        28        22        38
0.1149271 0.1795824 0.3288547 0.4677056
```

Formal assessment by dummy variables

- The following model adds indicator variables for six candidate observations.

```
u1=rep(0,n);u2=u3=u4=u5=u6=u1
u1[37]=1; u2[38]=1;u3[9]=1;u4[22]=1;u5[27]=1;u6[28]=1
glm(formula = log.survival ~ clotting + prognostic + enzyme +
liver + u1 + u2 + u3 + u4 + u5 + u6, data = surgical.dat)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.4019573  0.0356101  11.288 1.93e-14 ***
clotting     0.0761649  0.0039496  19.284  < 2e-16 ***
prognostic   0.0097609  0.0003329  29.324  < 2e-16 ***
enzyme       0.0096685  0.0002617  36.941  < 2e-16 ***
liver       -0.0003676  0.0064465  -0.057 0.954786
u1          -0.0811313  0.0323141  -2.511 0.015889 *
u2           0.1221444  0.0367687   3.322 0.001831 **
u3          -0.1057793  0.0310642  -3.405 0.001443 **
u4           0.1504064  0.0332193   4.528 4.67e-05 ***
u5           0.1215277  0.0307224   3.956 0.000281 ***
u6          -0.1060438  0.0357442  -2.967 0.004899 **
```

- With Bonferroni adjustment for six tests, all candidates except observation 37 remain significant at level 0.05.

Summary of the example

For the surgical unit data, diagnostics suggest

- no major systematic discrepancy in the mean structure,
- no strong evidence of heteroscedasticity,
- possible heavy tails in the residual distribution,
- several observations that merit attention under local diagnostics.

Final summary

Model diagnostics addresses two questions.

- Does the fitted model show systematic inadequacy?
- Are there individual observations with unusual leverage, residual, or influence?

Residual plots are the main tools for systematic diagnostics. Hat values, studentized deleted residuals, and Cook's distance are the main tools for local diagnostics.

Any remedy or model modification is a separate issue and will be discussed next.